Experiment – 3

Student Name: Shagun **UID: 24BDA70214**

Branch: CSE (Data Science) **Section/Group: 24BDS-SPE-1**

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Subject: ADBMS

AIM

To understand and implement SQL queries using Correlated Subqueries, JOIN operations, GROUP BY, and aggregate functions to retrieve and analyze data for solving real-world database problems.

Question 1

You are given an **Employee** table containing **EID, DEPT, and SCORES**. Write an SQL query to update/display the **highest score for each respective department**. Every employee in the same department should show the department's maximum score.

```
SELECT EID,
DEPT,
(
SELECT MAX(SCORES)
FROM Employee E2
WHERE E1.DEPT = E2.DEPT
) AS SCORES
FROM Employee E1;
```

	eid [PK] integer	dept character varying (10)	scores numeric
1	1	D1	5.28
2	2	D1	5.28
3	3	D1	5.28
4	4	D2	8.00
5	5	D1	5.28
6	6	D2	8.00
7	7	D3	9.00
8	8	D4	10.20

Question 2

SALES ANALYSIS PART 1

A company maintains records of products sold by different salespersons. The management wants to prepare a sales performance report by identifying the salesperson(s) who generated the highest overall sales revenue. If multiple salespersons have the same highest total sales amount, all of them must be included in the final report. Return the seller ID(s) satisfying this condition.

```
SELECT seller_id
FROM Orders
GROUP BY seller_id
HAVING SUM(amount) =
(
   SELECT MAX(max_amt)
   FROM
   (
      SELECT seller_id,
             SUM(amount) AS max_amt
      FROM Orders
      GROUP BY seller_id
   ) AS T
);
```

	seller_id integer
1	3

Question 3

A company maintains employee salary records for different departments. The management wants to prepare a report containing employees whose salaries fall among the **top three highest distinct salary values** within their respective departments. If multiple employees receive the same qualifying salary, all such employees must be included in the final report. Display the department name, employee name, and salary for every qualifying employee.

```
SELECT
   D.name AS Department,
   E1.name AS Employee,
   E1.salary AS Salary
FROM Employee E1
JOIN Department D
ON E1.departmentId = D.id
WHERE 3 >
(
   SELECT COUNT(DISTINCT E2.salary)
   FROM Employee E2
   WHERE E2.departmentId = E1.departmentId
   AND E2.salary > E1.salary
)
ORDER BY D.name, E1.salary DESC;
```

	department character varying (50)	employee character varying (50)	salary integer
1	IT	Max	90000
2	IT	Joe	85000
3	IT	Randy	85000
4	IT	Will	70000
5	Sales	Henry	80000
6	Sales	Sam	60000